Road factors and soil type: what really affects layer stiffness?

Backcalculated moduli (own temperature correction, compromise limits) · 6,667 locations on 16 roads · Östergötland 2022–2024

Summary

- **Soil type is the only clear, robust explanation.** Roads on all moraine types and on bedrock have a 34–45% stiffer subbase (E3) than roads on glacial clay (at least +15% in the strictest version). Bedrock also gives a stiffer base (E2, +47%) and subgrade (E4, +30%), sandy moraine a stiffer subgrade (+17%).
- **Soil does not affect the asphalt (E1).** An apparent +23% for moraine disappeared when comparing within the same road.
- **Heavy traffic** goes with a stiffer asphalt layer (+14% per doubling), also within the same road. It does not robustly affect the lower layers.
- **Ditch depth has no effect** on any layer. **ghum_max** has a small robust effect (−2% per 10 units on E2–E4), but its meaning is unknown.
- **These parameters explain only part of the stiffness** (R^2 12–48%). For the subgrade, 81% of all variation is between surveys at the same location (season, moisture), which no road parameter can explain.

Method

Each location gets one value per layer: the geometric mean of its usable surveys (fit within 5%, at least 2 surveys). One regression per layer: log(E) = soil type + layer thicknesses + $\log_2$(heavy traffic) + ditch depth + ghum_max. Soil types with fewer than 150 locations are grouped as “Other” (not interpreted). Neighbouring 20 m points are almost copies, so uncertainty is computed with 2 km road blocks as the unit (clustered standard errors).

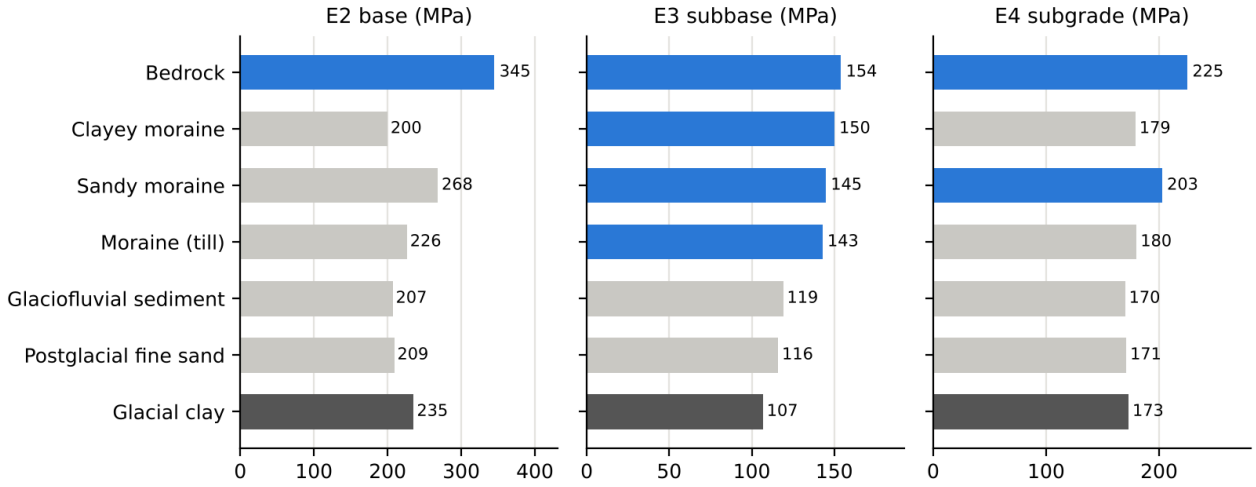
To make sure a result is not an artefact, every regression was run in **three versions**:

Version	What changes	What it protects against
A Main model	All locations	–
B Within the same road	Road added as a factor: locations are only compared with locations on the same road	A road that is just built better (age, standard) being mistaken for a soil or traffic effect
C Without limit values	Locations where this layer hit a modulus limit are removed	Values stuck at 10,000 or 1,000 MPa that are not true values

A result is called **robust** only if it is significant ($p < 0.05$) with the same sign in A, B and C. This matters because 52 effects are tested: at $p < 0.05$ two or three would look significant by chance. Layer thicknesses stay in the model as controls only: they are inputs of the backcalculation itself, so their effect is mostly a calculation effect (thicker layer → lower modulus for the same bowl) and is not interpreted. Heavy traffic is on a log scale (effect per doubling), because it ranges from about 30 to over 2,000 heavy vehicles/day.

1. Soil type

The figure shows the **adjusted typical value**: the stiffness each soil type would give if every location had the same thicknesses, traffic, ditch depth and ghum_max. Blue = robust difference from glacial clay; grey = not robust; dark grey = glacial clay.



Soil type (Swedish)	Loc.	E1 asphalt	E2 base	E3 subbase	E4 subgrade
Glacial clay (Glacial lera)	1,479	3,625 MPa	235 MPa	107 MPa	173 MPa
Moraine (till) (Morän)	1,371	+23%	-4%	**+34%** (+15% to +34%)	+4%
Sandy moraine (Sandig morän)	987	-2%	+14%	**+36%** (+21% to +37%)	**+17%** (+17% to +21%)
Glaciofluvial sediment (Isälvssediment)	895	-13%	-12%	+11%	-2%
Clayey moraine (Lerig morän)	287	-17%	-15%	**+40%** (+16% to +40%)	+3%
Bedrock (Urberg)	220	+9%	**+47%** (+32% to +54%)	**+45%** (+34% to +50%)	**+30%** (+30% to +32%)
Postglacial fine sand (Postglacial finsand)	214	-3%	-11%	+8%	-1%

First row: typical value on glacial clay. Other rows: % vs glacial clay (version A). **Bold** = robust, with the range over versions A–C in brackets; grey = not robust.

What did not hold up

- Moraine +23% on E1 (version A) dropped to +3% within the same road (B): it was a road effect, not a soil effect.
- Glaciofluvial sediment, clayey moraine and postglacial fine sand on E1/E2, and moraine on E4, changed sign or lost significance between versions.

2. Other road factors

Factor (one step)	Version	E1 asphalt	E2 base	E3 subbase	E4 subgrade
Asphalt thickness (+10 mm)	A	**-7%**	**-7%**	+0%	**-2%**
	B	**-8%**	**-6%**	+1%	**-1%**
	C	**-5%**	**-4%**	-0%	**-2%**
Base thickness (+100 mm)	A	+4%	**-9%**	**-5%**	**+6%**
	B	+2%	**-10%**	**-5%**	**+6%**
	C	+2%	**-10%**	-2%	**+6%**
Subbase thickness (+100 mm)	A	+2%	+1%	-0%	+1%
	B	+2%	+1%	-1%	+2%
	C	+1%	+1%	**-2%**	+1%
Heavy traffic (doubled)	A	**+14%**	**+8%**	**+4%**	**+2%**
	B	**+18%**	+7%	-0%	+1%
	C	**+7% R**	**+5%**	+2%	+2%
Ditch depth (+1 m)	A	+1%	+2%	+2%	+0%
	B	+2%	+1%	+2%	+0%
	C	-1%	+1%	+1%	+0%
ghum_max (+10)	A	-1%	**-2%**	**-2%**	**-2%**
	B	-0%	**-2%**	**-3%**	**-2%**
	C	0%	**-2% R**	**-2% R**	**-2% R**

Bold = significant, grey = not significant, **R** = robust. Thickness rows are controls only and not interpreted.

- **Heavy traffic:** doubling heavy traffic goes with +14% E1 (range +7% to +18%), also within the same road. Traffic does not make asphalt stiffer: busier stretches are likely built and maintained to a higher standard (newer or thicker asphalt mixes). On E2–E4 the traffic effect is not robust.
- **Ditch depth:** no effect on any layer in any version.
- **ghum_max:** −2% per 10 units on E2, E3 and E4 in all versions. Small, and its meaning must be confirmed with the data provider before it is interpreted.

3. How much do these parameters explain?

Layer	R^2 A	R^2 B (with road)	R^2 C	Variation between locations
E1 asphalt	36%	49%	23%	87%
E2 base	48%	55%	31%	64%
E3 subbase	12%	22%	8%	47%
E4 subgrade	12%	17%	12%	19%

- R^2 = the share of differences between locations explained by the parameters. It is moderate for E1/E2 and low for E3/E4.
- The last column shows why: for the subgrade, only 19% of the variation is between locations; 81% is between surveys at the same location (season and moisture). No fixed road parameter can explain that part. A low R^2 for E4 is therefore expected and not a sign of a bad analysis.

Limitations

- Soil type comes from a soil map (natural soil), not from samples under the road. The road embankment can be different material.
- This is an observational analysis: it shows what goes together, not what causes what.
- “Other soils” mixes 24 small soil groups (peat, silt, fill, …) and is not interpreted.

Code: soil_effect.py (all numbers, versions A–C, are written to data/soil_effect/regression_all_versions.csv). Results are also on the website, tab “Road factors & soil”.